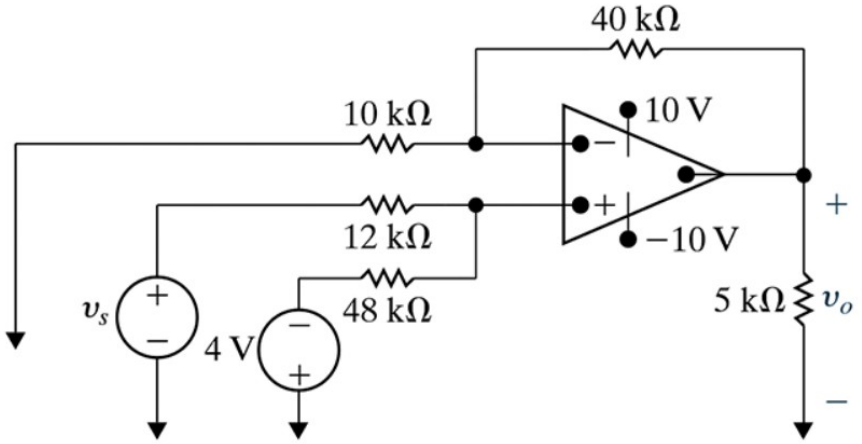
EE 2010
Fall 2026

Name Wyatt Evans

Game 24
10/2/2026

a) Find the gain $\left(\frac{v_o}{v_s}\right)$ for the circuit below using node voltage. Show detailed work.

b) Find the current coming out/going into the output pin of the op-amp under the following condition. You can use KCL at the output node to determine the current.
   i) $v_s = 1\ VDC$
   ii) $v_s = 2\ VDC$



$$0 = \frac{V_{in} - V_s}{12} + \frac{V_{in} - 4V}{48} = 5V_{in} - 4V_s - 4$$

$$0 = \frac{V_{in}}{10} + \frac{V_{in} - V_o}{40} = 5V_{in} - V_o$$

$$V_o - 4V_s - 4 = 0$$

$$V_o = 4V_s + 4V$$

gain

wei shen said to leave it like this for gain and not do v_out over v_in

$$i_o = \frac{V_o}{5K\Omega} = \frac{4(V_s) + 4}{5000}$$

$$= 1.6mA,\ 2.4mA$$